

QUANTUM KEY DISTRIBUTION

The Future Of Secure Communication

We all know that security of information is important, more so when the information concerns banking and financial services, military and defence networks, and various government offices spread all over the globe. The challenge would increase with the introduction of quantum computers that are expected sooner than later. So, let's find out what's being done



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Modern communication technologies and data security are heavily dependent on encryptions to maintain the secrecy of the information being transmitted or stored. In any cryptographic algorithm, information security depends entirely on the privacy of keys. Today's state-of-the-art public key infrastructure (PKI), which is mostly used across the globe to secure the internet, is based on mathematically complex algorithms. The security is based on the assumption that it is virtually impossible, even for the most advanced classical computer, to perform certain mathematical functions, like prime factorisation of a very large integer, in a reasonable time.

However, with the rapid advancements in quantum computing and quantum algorithms, this assumption does not hold very strong. Google's claim of quantum supremacy in October 2019, to perform a sequence of operations in 200 seconds that would take a classical supercomputer about thousands of years to complete, points toward the vulnerability of the existing cryptographic systems in front of an adversary having

access to a quantum computer.

Quantum key distribution (QKD) is a method of key exchange that eliminates any probability of eavesdropping. Fundamental rules of quantum mechanics ensure that any case of eavesdropping changes the system irreversibly, and the attempts of eavesdropping get detected.

Basics of cryptography

Cryptography is commonly used in communication networks to protect the data being transmitted from the hands of malicious or unwanted users. Before transmission, the information (plain text message) is encrypted to form a ciphertext that is unreadable without access to the corresponding decrypting key. At the receiver, the ciphertext is decrypted to get back the original plain text message. The process is depicted in Fig. 1.

Depending on the key used during the encryption and decryption processes, the cryptographic algorithms can be broadly classified into two categories:

Symmetric key cryptography. In this case, the same key is used for encryption as well as decryption. The key lengths are relatively smaller (128-256 bits). Examples: Advanced Encryption System (AES), Data Encryption Standard (DES).

Asymmetric/Public key cryptography. In this case, the keys used for the encryption and decryption process are different. A public key is used for encryption and a mathematically related private key is used for decryption. Data encrypted with a public key can only be decrypted with



Fig. 1: Process of encryption and decryption